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**AIM:** Implementation of Linear Regression Model

**Dataset:** housing-dataset

```
#Importing important Libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns
```

```
#downloading Dataset
import kagglehub
path = kagglehub.dataset_download("ashydv/housing-dataset")
```

```
Using Colab cache for faster access to the 'housing-dataset' dataset.
```

```
#printing the path
print(path)
```

```
/kaggle/input/housing-dataset
```

```
import os
os.listdir(path)
```

```
['Housing.csv']
```

```
#Loading and Reading DataSet
df=pd.read_csv(path+"/Housing.csv")
```

```
print(df.head())
print(df.info())
```

```
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
None
```

```
#Feature Selection
X=df[["area","bedrooms","bathrooms","stories","parking"]]
y=df["price"]
```

```python
#Training and Testing
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

```python
#Model Selection
model=LinearRegression()
```

```python
#Model Training
model.fit(X_train,y_train)
```

```
▾ LinearRegression ⓘ ?
LinearRegression()
```

```python
#Prediction
y_pred=model.predict(X_test)
```

```python
print("Model Coefficients:",model.coef_)
print("Model Intercept:",model.intercept_)
```

```
Model Coefficients: [3.08866956e+02 1.51246751e+05 1.18573171e+06 4.95100763e+05
 3.37660830e+05]
Model Intercept: 51999.67680883873
```

```python
# Print equation
print("Regression Equation:")
print("Price =", round(model.intercept_, 2), end=" ")

for coef, feature in zip(model.coef_, X.columns):
    print(f"+ ({round(coef,2)} * {feature})", end=" ")
```

```
Regression Equation:
Price = 51999.68 + (308.87 * area) + (151246.75 * bedrooms) + (1185731.71 * bathrooms) + (495100.76 * stories) + (337660.83 * parking)
```

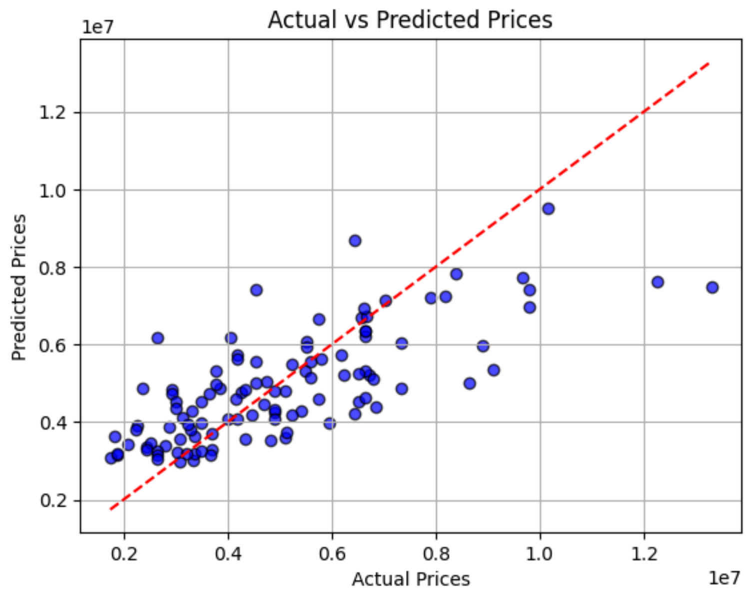
```python
#Evaluation
mse=mean_squared_error(y_test,y_pred)
rmse=mse**0.5
mae=mean_absolute_error(y_test,y_pred)
r2=r2_score(y_test,y_pred)
print(f"MSE:{mse:.2f}")
print(f"RMSE:{rmse:.2f}")
print(f"MAE:{mae:.2f}")
print(f"R2 Score:{r2:.2f}")
```

```
MSE:2292721545725.36
RMSE:1514173.55
MAE:1127483.35
R2 Score:0.55
```

```python
#Visuallization
plt.scatter(y_test, y_pred, color="blue", edgecolor="k", alpha=0.7)

plt.plot(
    [min(y_test), max(y_test)],
    [min(y_test), max(y_test)],
    color="red",
    linestyle="--"
)

plt.title("Actual vs Predicted Prices")
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.grid(True)
plt.show()
```



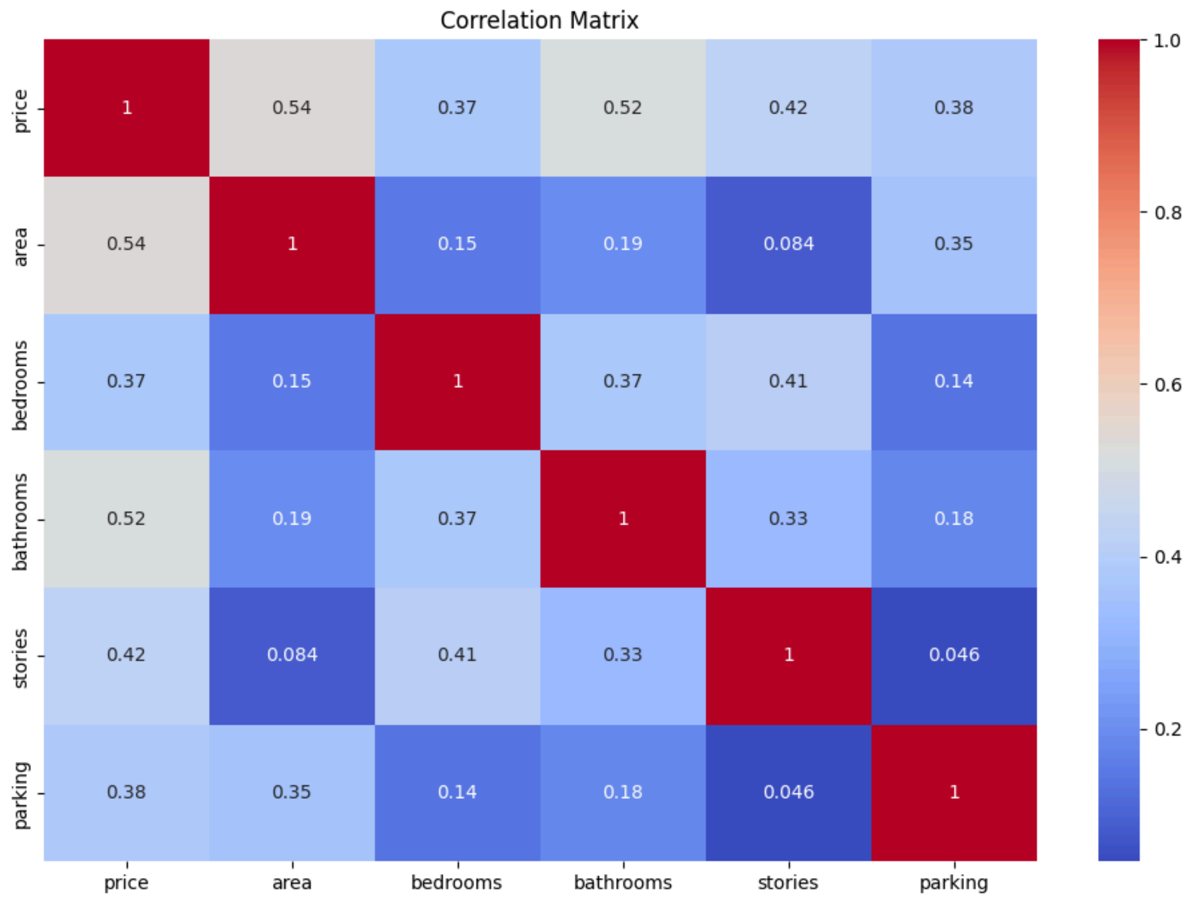

```python
print(X.columns)
```

```
Index(['area', 'bedrooms', 'bathrooms', 'stories', 'parking'], dtype='object')
```

```python
corr_matrix = df.corr(numeric_only=True)
print(corr_matrix)
```

```
              price      area  bedrooms  bathrooms   stories   parking
price      1.000000  0.535997  0.366494   0.517545  0.420712  0.384394
area       0.535997  1.000000  0.151858   0.193820  0.083996  0.352980
bedrooms   0.366494  0.151858  1.000000   0.373930  0.408564  0.139270
bathrooms  0.517545  0.193820  0.373930   1.000000  0.326165  0.177496
stories    0.420712  0.083996  0.408564   0.326165  1.000000  0.045547
parking    0.384394  0.352980  0.139270   0.177496  0.045547  1.000000
```

```python
plt.figure(figsize=(12,8))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```




```python
print(df.corr(numeric_only=True)["price"].sort_values(ascending=False))
```

```
price        1.000000
area         0.535997
bathrooms    0.517545
stories      0.420712
parking      0.384394
bedrooms     0.366494
Name: price, dtype: float64
```

```python
#Testing on New Data
new_house = pd.DataFrame([{
    "area": 3000,
    "bedrooms": 3,
    "bathrooms": 2,
    "stories": 2,
    "parking": 1
}])
```

```python
predicted_price = model.predict(new_house)

print("Predicted House Price:", round(predicted_price[0], 2))
```

```
Predicted House Price: 5131666.58
```